

Factoring Special Forms in Area and Volume Models

Differences of squares, perfect-square trinomials, and pulling out the common factor first

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- Every expression below measures a real area or volume, so keep the units in mind as you work: they tell you whether your factors are lengths.
 - Factor **completely**. Always check for a common factor before you reach for a special pattern.
 - Two patterns do all the work here: $a^2 - b^2 = (a - b)(a + b)$ and $a^2 \pm 2ab + b^2 = (a \pm b)^2$.
 - Show your work in the space provided, then write your final answer on the answer line.
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1. A square sheet of copper has area x^2 square centimetres. A square hole of area 36 square centimetres is punched out of its middle, leaving $x^2 - 36$ square centimetres of metal. Write the remaining area as a product of two binomials.

Answer: _____

2. A square patio is tiled so that its area is $x^2 + 10x + 25$ square metres. Factor the area completely so that the side length of the patio, in metres, can be read from the answer.

Answer: _____

3. A square banner measures $3x$ centimetres on each side. A square logo measuring 7 centimetres on each side is cut out of it, leaving $9x^2 - 49$ square centimetres of fabric. Write the remaining area as a product of two binomials.

Answer: _____

4. The floor of a square storage shed has area $4x^2 + 12x + 9$ square feet. Factor the area completely so that the side length of the shed, in feet, can be read from the answer.

Answer: _____

5. A rectangular deck has area $2x^2 - 50$ square feet. Factor the expression completely. Take out the common factor first.

Answer: _____

6. A rectangular window pane measures $(x + 7)$ inches along one edge and $(x - 7)$ inches along the other. Write its area, in square inches, in standard form with no parentheses.

Answer: _____

7. Two square vegetable plots lie side by side. The larger plot has side $5x$ metres and the smaller plot has side $4y$ metres, so the difference of their areas is $25x^2 - 16y^2$ square metres. Write that difference as a product of two binomials.

Answer: _____

8. A square rug and its carpet border together cover $3x^2 - 30x + 75$ square feet. Factor the expression completely: take out the greatest common factor first, then look at what is left.

Answer: _____

9. The volume of a shipping crate, in cubic inches, is modelled by $x^3 - 16x$, where x is a length in inches. Factor the expression completely so that the three dimensions of the crate can be read from the answer.

Answer: _____

10. A square window pane has area $16x^2 - 24x + 9$ square inches.

- (a) Factor the area completely, and write the factored form on the answer line.
- (b) In the space above, explain how the numbers 16, 24 and 9 tell you this expression is a perfect-square trinomial rather than a difference of squares.

Answer: _____